Jay Hemnani

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Summary

Data analyst. Most recently built ETL pipelines, Tableau and Power BI dashboards, and demand forecasting models for a multi-property hotel group. Works in SQL, Python and R.

Technical Skills

Programming: Python, SQL (Advanced), R, JavaScript
Analytics & Statistics: Pandas, NumPy, Scikit-learn, Statistical Analysis, A/B Testing, Hypothesis Testing, Regression, Cohort Analysis
Visualization & BI: Tableau, Power BI, Plotly, Streamlit, Dash, Excel (Advanced), Google Sheets, Data Storytelling
Databases: PostgreSQL, MySQL, MongoDB, BigQuery, Data Warehousing, Star Schema Design
Data Processing: ETL/ELT Pipelines, Data Cleaning, Data Validation, Apache Airflow, dbt
Tools: Git, Jupyter Notebooks, Google Analytics, Mixpanel, Looker

Experience

Data Analyst, *Elite Hotel Group* Summer 2025 | Remote

- Defined **metrics and SLAs** with finance and operations, which became the spec for the ETL and forecasting work.
- Built automated ETL pipelines in SQL and Python, cutting manual data preparation by **40%** across multi-property analytics.
- Built dashboards in **Tableau and Power BI** for occupancy, revenue, and KPI tracking across multiple hotel properties.
- Built **time-series demand forecasting models** for dynamic pricing and revenue management decisions.

Technical Consultant, *Independent* 2022 – 2024 | Remote

- Did data analytics and pipeline consulting for small businesses; built their **reporting automation** and data infrastructure.
- Designed **A/B testing frameworks** and tracked engagement metrics to improve client campaigns and product decisions.

AI/ML Intern, *Amnex* Jan – May 2022 | Gujarat, India

- Built a credit fraud detection ensemble (Random Forest, XGBoost, SMOTE for class imbalance); **94% precision** on transaction data.
- Built **automated analytics dashboards** on statistical models for operational reporting and anomaly detection.

Software Engineering Intern, *Cactus Creatives* May – Nov 2019 | Gujarat, India

- Built a cloud-native communication platform for first responders on Azure; set up CI/CD pipelines that cut deployment time by **60%**.

Projects

Metabolic Syndrome Risk Validation | *R, Python, survival, XGBoost*

- Tested 2 published metabolic-syndrome risk scores on a **17,031-adult NHANES cohort** linked to US mortality data, against Framingham, ACC/AHA PCE and FINDRISC.
- Survey-weighted Fine-Gray and Cox models, time-dependent AUC and a 500-replicate bootstrap. The result was mixed: for cardiovascular mortality Framingham reached **AUC 0.858** against 0.660 for RMRS; for diabetes mortality RMRS ran close to FINDRISC (0.752 vs 0.770).
- 15-step R/Python pipeline with a written protocol and pinned dependencies; one Makefile rebuilds every table from raw NHANES.

Stock Data Platform | *Python, SQL, Kafka, Airflow, TimescaleDB, Chart.js*

- Designed a **10-table star schema** (3 dimension, 7 fact) for prices, fundamentals, earnings, filings and macro data on 10 US stocks.
- **23 Airflow DAGs** load 4 sources (yfinance, SEC EDGAR, FRED, Alpaca) incrementally from the last loaded date, so reruns are safe.
- Built a Chart.js dashboard for price trends, fundamentals, earnings surprises and macro indicators.

Open Source Contributions

550+ merged pull requests across 60 open-source projects, including vLLM, Effect, Caddy, Lightpanda, radare2, DSPy, Home Assistant and Crawlee.
A plain-English note on each one: jayhemnani.in/oss

Education

Pandit Deendayal Energy University (PDEU) 2018 – 2022 | Gandhinagar, India
B.Tech in Computer Engineering | *GPA: 8.7/10*

- **Coursework:** Big Data Analytics, Database Systems, Natural Language Processing, Computer Vision.